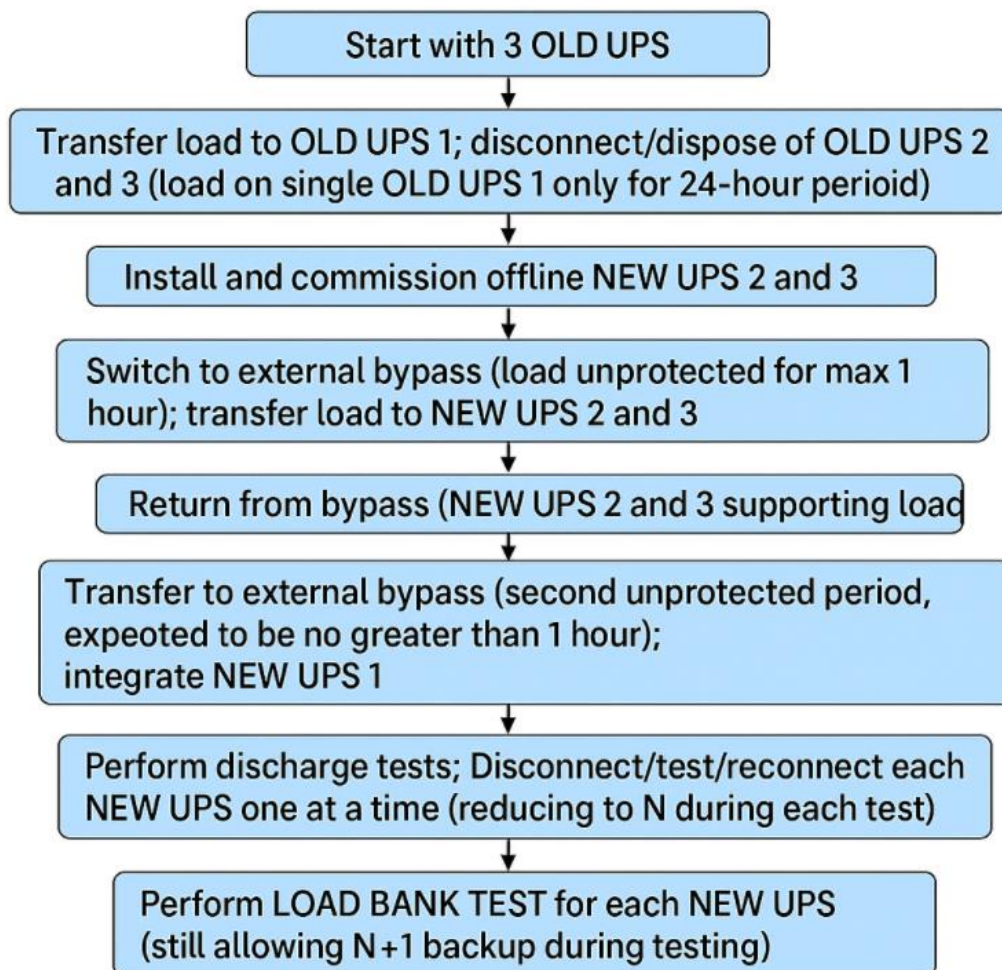


Forth Valley – UPS installation sequence of work proposed by ABB



Key Concerns

Extended Operation on a Single Old UPS (24-hour duration – Concern: High Risk)

- **Issue:** The proposed system runs without redundancy (N only) on an aging OLD UPS 1 for 24 hours.
- **Risks:**

- **Patient Harm in Critical Areas:** If OLD UPS 1 fails (e.g., due to overload, battery depletion, or internal fault), the load shifts to generators (10–15 seconds delay). In neonatal/ICU, even a brief interruption risks ventilator stoppage, incubator cooling, or monitor failure, potentially causing hypoxia, brain injury, or death in ventilated infants/adults. Children's ward equipment (e.g., infusion pumps) could disrupt, leading to medication errors.
- **Probability:** due to the 24-hour limit—mains/UPS failure chance ~0.1–1% per hour in reliable UK grids. However, OLD UPS 1's age increases fault likelihood (e.g., capacitor wear or battery degradation, common in end-of-life systems).
- **Secondary Impacts:** Potential equipment damage from transients; staff response challenges in high-acuity areas. Short duration reduces cumulative exposure but doesn't eliminate the single point of failure.
- **Non-Compliance:** Deviates from SHTM 06-01 Grade A requirements (N+1 redundancy for no-break power <0.5s interruption). Acceptable for very brief maintenance in non-critical settings, but not ideal for FVRH's life-dependent loads.
- **Recommendation for Board:** Mandate pre-works testing of OLD UPS 1 (load bank verification) and contingencies like portable UPS or enhanced generator readiness. If possible, shorten further or eliminate via modular install.

Unprotected External Bypass Periods (Up to 1 Hour Each – Concern: High Risk)

- **Issue:** Two bypass switches: First (after commissioning NEW 2/3) for load transfer; second for integrating NEW 1. During these, load is unprotected by UPS, relying solely on mains (with generator backup after 10–15s).
- **Risks:**
 - **Even brief mains dips (common in grids) cause instant outages.** In ICU/neonatal, 10–15s delays risk asphyxiation (ventilators stop) or organ damage; children's ward alarms fail, missing deteriorations.
 - **Cumulative exposure:** Two periods double the window for incidents, especially if delayed by issues (e.g., commissioning faults).
 - **Patient Harm:** Potential fatalities/multiple casualties, e.g., similar global hospital outages have led to NICU deaths from equipment failure.
- **Non-Compliance:** Exceeds SHTM 06-01 no-break tolerances; not aligned with NHS Zero Harm goals.

- **Recommendation for Board:** Proceed as suggested but with caution. Liaise with SSE to confirm grid stability with no planned outages at specific BYPASS times. External BYPASS is also weather dependant.